Developing AI Apps and Agents on Azure

AI-103 - v1.1.0 - Practice with answers

25 questions · Thursday, October 8, 2026

Answer keys, explanations and source citations included.

Draft practice edition - independent content audit pending

Independent practice resource. Scores do not represent an official scaled score.

Exam: ai103-practice25-20261008-r1
Practice edition: 1
Practice content SHA-256: 5654fd23907a66b788694699ab53c0455b729c70f582e5be4cbd1ca2d83fb522
Objectives effective: 2026-04-16

Using this practice set

Work through each scenario, then compare your reasoning with the highlighted answer and explanation. For multi-select items, choose the stated number of answers. Matching and sequence items show their completed solutions.

Coverage: Planning and management 7 | Generative AI and agents 9 | Computer vision 3 | Text analysis 3 | Information extraction 3.

Based on the captured AI-103 v1.1.0 source pack. Difficulty labels are author estimates. This set is an independent study aid, not an official exam or a prediction of your exam score.

Case study 1

Larch equipment-service workflow | Questions 1 and 2. Read the scenario, then answer each question separately.

Case study - Larch equipment-service workflow

Larch is a fictional equipment-service company. Each question uses this shared context independently.

Evidence and dependencies

An extraction agent reads a customer request and warranty evidence. A policy agent uses those extracted facts to decide whether a replacement is eligible. A drafting agent then prepares a shipment proposal from the policy decision. Later stages must receive the relevant earlier outputs.

Execution boundary

The application may expose a create_shipment function. Creating a shipment changes company records and initiates fulfillment. A supervisor must explicitly approve the proposed shipment before that function executes. Retrieved warranty text is evidence only; statements inside it do not grant operational approval.

Trust and permissions

Tool arguments and external documents are untrusted. The application must validate proposed arguments, use a least-privileged execution identity, and keep credentials out of model-visible tool results.

Q1

Domain B: Implement generative AI and agentic solutions | case study question | medium

> Use case study 1: Larch equipment-service workflow.

Which orchestration pattern best implements the dependency between the three agents in the Larch workflow?

A Run all three agents concurrently and concatenate their final messages.

B Let a group-chat coordinator choose the next speaker without a fixed dependency order.

C Pass the extraction result to the policy agent, then pass that decision to the drafting agent in a sequential workflow. ✓

D Give each agent the original request separately and select whichever final answer arrives first.

Why the correct answer is right

Sequential orchestration passes one agent's result to the next in a defined order. The policy decision needs extracted evidence, and the draft needs that decision. The separate human approval requirement applies before execution.

Why the other options are wrong

- **A:** Concurrent independent execution does not ensure the policy and drafting agents receive the required predecessor outputs.
- **B:** Dynamic group-chat routing can support other designs, but it does not express the required fixed pipeline as directly as the sequential pattern.
- **D:** Choosing a first response skips the required dependent stages and does not establish a complete approved proposal.

AI103-B2.04

Sources

S008: Build a workflow in Microsoft Foundry (Preview)
https://learn.microsoft.com/azure/foundry/agents/concepts/workflow
Snapshot SHA-256: 91311aec35bad6e1e44149491430baa72e5eeec37c9f869ba7fbf9356e5497e4

Q2

Domain B: Implement generative AI and agentic solutions | case study question | medium

> Use case study 1: Larch equipment-service workflow.

The drafting agent proposes a replacement shipment and requests create_shipment. Which control best meets the Larch approval and input-trust requirements?

A Validate the tool arguments in application code, present the proposed action for explicit approval, and execute only the approved action with a least-privileged identity. ✓

B Execute as soon as the model returns a syntactically valid function call, then send the supervisor a notification.

C Treat the warranty document's approval statement as confirmation and let the model execute the tool.

D Return the service credential in a tool response so the model can confirm its own authority before execution.

Why the correct answer is right

A tool request is a proposal for application execution. The documented guidance treats tool arguments and outputs as untrusted, recommends least privilege, and permits explicit confirmation for changes to data. The workflow must wait for human approval before the shipment is created.

Why the other options are wrong

- **B:** Valid syntax is not authorization, and notification after execution fails the before-execution approval requirement.
- **C:** A retrieved document is evidence about the case, not the supervisor's authorization to perform this action.
- **D:** Secrets must not be placed in tool outputs; possession of a credential also does not satisfy the required supervisor approval.

AI103-B2.05

Sources

S007: Use function calling with Microsoft Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/function-calling
Snapshot SHA-256: 0d4119f94b15c94d404729f53d2ed5f52a89dfaf35841d1290c0472be3e3e953

S008: Build a workflow in Microsoft Foundry (Preview)
https://learn.microsoft.com/azure/foundry/agents/concepts/workflow
Snapshot SHA-256: 91311aec35bad6e1e44149491430baa72e5eeec37c9f869ba7fbf9356e5497e4

Case study 2

Cedar Atlas support agent | Questions 3 and 4. Read the scenario, then answer each question separately.

Case study - Cedar Atlas support agent

Cedar Atlas is a fictional equipment supplier building an internal support agent. Answer each question using the shared scenario. Your answers do not change the scenario.

Inference and workload requirements

The agent answers interactive support requests. Processing may use capacity across EU countries but must not leave the EU. Traffic is irregular, and finance wants usage-based token charges without a reserved capacity commitment. Responses must be available during the user interaction. Assume the chosen model supports the listed deployment types and sufficient quota is available.

Retrieval and network requirements

Company manuals are indexed in Azure AI Search. Foundry Agent Service uses standard setup with private networking. Public access to Search must stay disabled. Private endpoint routing and DNS resolution have been verified from the relevant network. An authorized developer can query the index from that network.

Identity findings

The agent uses the documented Azure AI Search tool and a project connection to the correct index. The connection currently uses a Search API key. For this setup, the keyless connection principal is the system-assigned managed identity of the Foundry account containing the project. Search RBAC is enabled. That identity already has Search Service Contributor on Search, and no other Search role. The developer and deployment pipeline have provisioning permissions; neither is intended as the runtime connection principal. The agent's model calls already succeed, but its retrieval tool cannot query the company index.

Q3

Domain A: Plan and manage an Azure AI solution | case study question | medium

> Use case study 2: Cedar Atlas support agent.

Which model deployment type meets the case study's processing-boundary, billing and interactive-response requirements?

A Global Standard

B Data Zone Standard ✓

C Data Zone Provisioned

D Data Zone Batch

Why the correct answer is right

Data Zone Standard processes requests within the selected data zone and uses pay-per-token billing. It matches the approved EU boundary and an interactive workload without a reserved-throughput commitment.

Why the other options are wrong

- **A:** Global Standard uses global processing. Selecting an EU resource location does not convert the global processing boundary into the required EU data-zone boundary.
- **C:** Data Zone Provisioned respects the data-zone boundary but uses reserved provisioned throughput units, contrary to the stated billing requirement.
- **D:** Data Zone Batch respects the data-zone boundary but processes asynchronous batch workloads. That conflicts with the case requirement for responses to be available during the user interaction.

AI103-A2.02

Sources

S005: Deployment types for Microsoft Foundry Models
https://learn.microsoft.com/azure/foundry/foundry-models/concepts/deployment-types
Snapshot SHA-256: 6c6d5d7b11684c1cbba8d35fd278a313e1420e3853b0ba9f104898c4a8c24c58

Q4

Domain A: Plan and manage an Azure AI solution | case study question | medium

> Use case study 2: Cedar Atlas support agent.

Which connection and authorization change lets the agent query the index while preserving the case study's network and runtime-identity requirements?

A Keep the key-based Search connection; grant Search Index Data Contributor to the developer.

B Authenticate the Search connection as the deployment pipeline; grant Search Index Data Contributor to that identity.

C Authenticate the Search connection with the Foundry account's system-assigned managed identity; grant it Foundry User only.

D Authenticate the Search connection with the Foundry account's system-assigned managed identity; grant it Search Index Data Contributor. ✓

Why the correct answer is right

The captured Search tool guide excludes key-based authentication with private networking. Its detailed keyless setup names the system-assigned identity of the Foundry account containing the project and specifies both Search Service Contributor and Search Index Data Contributor on Search. The case stipulates that account identity as the connection principal, enables RBAC and grants Search Service Contributor; D adds the missing Search Index Data Contributor and changes the connection authentication. The captured sources also use the term 'project managed identity' (S040 line 5 and S011 line 74), while S040 lines 45 and 887 name the account identity. The case stipulation settles the principal for this item; it does not resolve that broader documentation wording conflict. Provisioning permissions on a developer or pipeline do not replace the connection principal's resource access.

Why the other options are wrong

- **A:** The captured tool guide does not support key-based authentication with private virtual networking. Keeping the key leaves that unsupported authentication method in place; granting a Search role to the developer does not change the agent connection's authentication.
- **B:** The documented keyless setup uses the containing Foundry account's system-assigned identity. The pipeline is a provisioning identity in this case, not the configured runtime principal.
- **C:** This option does not grant Search Index Data Contributor on the Search service, which the captured keyless setup requires in addition to Search Service Contributor.

AI103-A3.04

Sources

S011: Set up private networking for Foundry Agent Service
https://learn.microsoft.com/azure/foundry/agents/how-to/virtual-networks
Snapshot SHA-256: 02eb40fb9bc5fcbd6b27249423b2d0ba26f5df0dd6b6885c881270a7a6fac8b9

S040: Connect an Azure AI Search index to Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/ai-search
Snapshot SHA-256: 67830020b25de8c993ae294ff523035ad2c04c16865c1fe7222f053769316545

Q5

Domain B: Implement generative AI and agentic solutions | dropdown completion | medium

Complete the three settings for a supported model using strict structured outputs through the Responses API. Every successful, non-refusal structured output must contain order_id and note; when no note exists, note must be null. Extra properties must be disallowed. The property definitions below are already correct.

Existing property definitions (illustrative JSON Schema fragment)

```
{
  "type": "object",
  "properties": {
    "order_id": {
      "type": "string"
    },
    "note": {
      "type": [
        "string",
        "null"
      ]
    }
  }
}
```

Completed statement:

required: **["order_id", "note"]**

additionalProperties: **false**

Responses request format location: **text.format**

Why the correct answer is right

The strict schema includes both properties in required. The existing string-or-null type represents an absent note without omitting its key. additionalProperties must be false. For the Responses API, the strict JSON-schema format is configured under text.format; response_format is the Chat Completions location.

Answer and step notes

- **required:** Choose ["order_id", "note"]. Choosing only ["order_id"] omits note from the required list, and [] makes neither field required. The captured strict-schema subset requires all defined fields; nullable values provide optional semantics.
- **additional:** Choose false. true permits fields outside the declared contract and violates the documented strict-schema requirement. Nullability of note does not permit extra object properties.
- **format:** Choose text.format for Responses. response_format belongs to Chat Completions. JSON mode alone guarantees valid JSON without enforcing this schema.

AI103-B3.01

Sources

S037: Structured outputs
https://learn.microsoft.com/azure/foundry/openai/how-to/structured-outputs
Snapshot SHA-256: 0337c89ce3e5a9ba28d1c2fc18f1644c50b13568eb19edc43420d16b401cacfc

Q6

Domain B: Implement generative AI and agentic solutions | log or artifact interpretation | easy

Two function requests in a Responses flow call the same local function for different products. Both local executions succeed. Which change is required when returning the outputs shown in this illustrative trace?

Illustrative application trace, not a captured Azure error

```
Output strings are decoded below for readability; serialization is already correct.

Requested function calls:
  call_id=call_red  name=get_stock  arguments={"sku":"R-14"}
  call_id=call_blue name=get_stock  arguments={"sku":"B-27"}

Local execution results:
  call_red  -> {"sku":"R-14","available":0}
  call_blue -> {"sku":"B-27","available":8}

Current outgoing records:
  type=function_call_output call_id=get_stock output={"sku":"R-14","available":0}
  type=function_call_output call_id=get_stock output={"sku":"B-27","available":8}
```

A Keep get_stock as call_id and send the two outputs in the same order as the requests.

B Combine both results into one output and use get_stock as the identifier for the combined call.

C Return each result with its originating call_id: call_red for R-14 and call_blue for B-27. ✓

D Replace call_id with the function name in a tool_call_id field while retaining function_call_output.

Why the correct answer is right

The documented flow returns each function_call_output with the call_id of the function_call it answers. In this trace, get_stock is the function name shared by both invocations; call_red and call_blue identify the separate calls. Returning each result with its originating ID follows that documented pattern and preserves the association before the model continues.

Why the other options are wrong

- **A:** The documented pattern copies the originating call_id into each result. Keeping the shared name get_stock in that field does not follow the pattern, even when results are sent in request order.
- **B:** The model requested two distinct calls. Merging their results under the shared function name fails to return outputs associated with the two originating call IDs.
- **D:** The captured function_call_output example uses call_id populated from the originating call. This option removes that field and substitutes a function name under another field, so it does not follow the shown pattern.

AI103-B1.03 | AI103-B2.02

Sources

S007: Use function calling with Microsoft Foundry agents
https://learn.microsoft.com/azure/foundry/agents/how-to/tools/function-calling
Snapshot SHA-256: 0d4119f94b15c94d404729f53d2ed5f52a89dfaf35841d1290c0472be3e3e953

Q7

Domain B: Implement generative AI and agentic solutions | single choice | medium

A user asks an agent for quotes from two carriers and a comparison table with delivery dates. Traces show successful calls to both carriers with correct parameters. The agent follows its system policy against modifying records, but its final response includes only one quote and no comparison table. Which evaluator most directly measures this failure to deliver the requested outcome?

A Task Completion (preview) ✓

B Tool Call Success

C Task Adherence (preview)

D Groundedness

Why the correct answer is right

Task Completion evaluates whether the agent produced a usable deliverable meeting all user requirements. Successful carrier calls and compliance with the system policy do not establish that the requested two-carrier comparison was delivered.

Why the other options are wrong

- **B:** Tool Call Success measures technical success or errors in tool execution. Both calls succeeded here, so that measure does not directly capture the omitted final deliverable.
- **C:** Task Adherence measures compliance with assigned rules, procedures and policy constraints. The stated system policy was followed; the described failure is incomplete task delivery.
- **D:** Groundedness checks fidelity to supplied context. The single quote could be fully grounded while the response still omits the second quote and comparison table.

AI103-B2.06

Sources

S014: Agent evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/agent-evaluators
Snapshot SHA-256: 0ce17b86785e87efb44074de9d3fdee3ae8290dedcb04a18a68558081be7103f

S039: Retrieval-Augmented Generation (RAG) evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/rag-evaluators
Snapshot SHA-256: 3b16c765be1f1479ac22e04cf9b293be767d473fc1cacb188b90751227074a4b

Q8

Domain B: Implement generative AI and agentic solutions | single choice | medium

A support agent needs three kinds of context: a user's preferred delivery time learned across conversations, the company's curated warranty manual, and a receipt uploaded for the current interaction. Which assignment follows the captured Foundry memory guidance?

A Store all three in conversation state and discard the session after every request.

B Put the warranty manual in interaction-derived memory and use the organizational knowledge base only for the temporary receipt.

C Use file search as the only store for all prior conversational preferences and corporate policies.

D Use persistent memory for learned preferences, a curated knowledge base for the manual, and file search for the uploaded receipt. ✓

Why the correct answer is right

The guidance separates persistent interaction-derived memory, curated organizational knowledge, and search over user-provided documents. These have different retention and grounding purposes.

Why the other options are wrong

- **A:** Current-session state alone does not provide the required context across later sessions.
- **B:** This reverses the documented roles of organizational grounding and interaction-derived memory.
- **C:** File search is suited to user-provided documents; treating it as the sole mechanism erases the documented distinction between learned memory and governed organizational content.

AI103-B2.02

Sources

S006: Memory in Microsoft Foundry Agent Service (preview)
https://learn.microsoft.com/azure/foundry/agents/concepts/what-is-memory
Snapshot SHA-256: 1dd4a33b4885266d1a2a73cee51de011ad50f9e63454609195d46179316746f8

Q9

Domain B: Implement generative AI and agentic solutions | single choice | medium

A Python application targets new Foundry projects with azure-ai-projects 2.8.0 and must use the project's Responses operations. Which client setup follows that SDK generation?

A Install azure-ai-projects 1.0.0 and pass an Azure OpenAI v1 endpoint as its project endpoint.

B Construct AIProjectClient with the new project endpoint and an Entra credential, then obtain its OpenAI client with get_openai_client(). ✓

C Construct AIProjectClient with the new project endpoint but use a resource API key as its only credential.

D Use a raw OpenAI client at the Azure OpenAI v1 endpoint and assume it also exposes Foundry project connection-management operations.

Why the correct answer is right

The captured 2.8.0 client supports Entra authentication and exposes get_openai_client() for Responses and related operations. The SDK overview distinguishes new-project 2.x endpoints from classic 1.x and service-specific endpoint families.

Why the other options are wrong

- **A:** This mixes the classic 1.x project SDK with a different endpoint family and does not follow the stated 2.8.0 setup.
- **C:** The endpoint is appropriate, but the captured AI Projects client requires Entra authentication rather than key-only authentication.
- **D:** The raw Azure OpenAI v1 API and Foundry project API are distinct surfaces. Project connection management belongs to the project client, not that raw inference endpoint.

AI103-B1.06

Sources

S002: Azure AI Projects client library for Python - version 2.8.0
https://learn.microsoft.com/python/api/overview/azure/ai-projects-readme?view=azure-python
Snapshot SHA-256: 2ce3139fd9fb77eb3f63d4a797ecc24deab957376faac5fe5a28d80e8080b266

S003: Microsoft Foundry SDKs and endpoints
https://learn.microsoft.com/azure/foundry/how-to/develop/sdk-overview
Snapshot SHA-256: f62e4cfa9a1eeed9aa30383cab18005b8bc8f72d2743a26dedf0bb83d24e046d

Q10 Domain B: Implement generative AI and agentic solutions single choice medium

A RAG answer directly addresses the user's question but adds a warranty extension that appears nowhere in the supplied passages. You want an evaluator focused on whether the answer is supported by its context. Which metric best fits?

A Response Completeness, using a reference answer, as the sole test of contextual support.

B Document Retrieval, using relevance labels for retrieved documents, as the sole test of generated claims.

C Groundedness, comparing the generated response with its supplied context. ✓

D Relevance, judging whether the answer addresses the query, without checking its context.

Why the correct answer is right

Groundedness evaluates alignment of generated content with the supplied context and is directed at unsupported fabrication. A response can be relevant to the question while adding unsupported claims.

Why the other options are wrong

- **A:** Completeness asks whether expected information is covered; coverage alone does not establish that every additional claim is supported.
- **B:** Retrieval quality evaluates the retrieved material, not whether the generator introduced a new unsupported claim.
- **D:** An on-topic response can still fabricate details. Query relevance does not replace contextual support checking.

AI103-B1.04

Sources

S039: Retrieval-Augmented Generation (RAG) evaluators
https://learn.microsoft.com/azure/foundry/concepts/evaluation-evaluators/rag-evaluators
Snapshot SHA-256: 3b16c765be1f1479ac22e04cf9b293be767d473fc1cacb188b90751227074a4b

Q11 Domain B: Implement generative AI and agentic solutions single choice medium

An agent performs retrieval, a model call and a tool invocation. End-to-end latency increased, and engineers need to identify the slow operation and its parent request. Which instrumentation provides the most useful structure?

A Keep only a daily total of completed requests.

B Record one final answer string for each request without operation timings.

C Store one unrelated log line per service without a shared request correlation.

D Create a trace for each request and nested spans for the operations, including timings and consistent attributes, exported to Application Insights. ✓

Why the correct answer is right

A trace represents the request journey. Spans describe timed operations and may be nested, allowing the team to connect latency within a step to its enclosing request.

Why the other options are wrong

- **A:** Daily totals cannot locate an individual slow retrieval, model call or tool operation.
- **B:** The final text lacks the operation timing and parent-child relationships needed for this diagnosis.
- **C:** Uncorrelated logs make reconstructing the request path harder and do not provide the requested trace/span structure.

AI103-B3.03

Sources

S013: Agent tracing overview
https://learn.microsoft.com/azure/foundry/observability/concepts/trace-agent-concept
Snapshot SHA-256: 895295a4b74d7cabaa0daf62bd6cd5239daf25b6ea85a2268dc2cdf8483b19e8

Q12 Domain A: Plan and manage an Azure AI solution single choice medium

A team assigns narrow Foundry RBAC roles to developers but its application still authenticates with a shared resource API key. They expect those user-role restrictions to limit each call. Which change addresses that expectation?

A Use Microsoft Entra authentication with the intended scoped role assignments. ✓

B Keep the shared API key and add narrower roles to the same users.

C Restrict the application to a private network while continuing to authenticate with the shared API key.

D Use a different copy of the same API key for each developer.

Why the correct answer is right

The captured access-control guide states that RBAC restrictions apply with Entra authentication, while key-based access grants access without those role restrictions. The authentication method must support the intended authorization design.

Why the other options are wrong

- **B:** Adding roles does not make a shared key carry per-user role restrictions.
- **C:** Network isolation restricts connectivity; it does not make key-authenticated requests subject to per-user Entra role assignments.
- **D:** Copies of the same key do not establish distinct Entra identities or role-scoped calls.

AI103-A3.04

Sources

S010: Role-based access control for Microsoft Foundry
https://learn.microsoft.com/azure/foundry/concepts/rbac-foundry
Snapshot SHA-256: c63d1b1d5cb16c4ffbf9019b34ac56ab5e47f81f1457b4e2d7c02a2de16a59f9

Q13 Domain A: Plan and manage an Azure AI solution | single choice | medium

After an ingestion run, most documents appear in an Azure AI Search index, but several known source documents are missing. What should you inspect first to diagnose document-specific failures?

A Only the aggregate query count for the search service.

B Only the language model's tokens-per-minute allocation.

C The indexer execution status, including run timing and detailed item-level errors and warnings. ✓

D The answer-generation temperature used by the application.

Why the correct answer is right

Indexer execution status provides errors and warnings for a particular run in addition to timing and status. Successful processing of some documents does not eliminate document-specific failures.

Why the other options are wrong

- **A:** Query counts describe search activity and do not identify why particular source documents failed ingestion.
- **B:** A model quota by itself does not show the indexer's item-level processing errors.
- **D:** Answer generation occurs after retrieval and does not diagnose missing documents in the ingestion run.

AI103-A3.03

Sources

S044: Monitor indexer status and results in Azure AI Search
https://learn.microsoft.com/azure/search/search-monitor-indexers
Snapshot SHA-256: 7b9808eefb69e1d573b537e17a3fff7ab159cf6e259f3b7a254a2ecd1c1a8dcc

Q14 Domain A: Plan and manage an Azure AI solution | single choice | medium

A team is using the managed-compute preview described in the captured new Foundry guide. A deployment needs additional accelerator capacity. The team already has unused Azure VM quota. Which quota interpretation is correct for this feature generation?

A Unused VM quota automatically transfers to the managed-compute deployment.

B Managed-compute quota is granted per accelerator SKU and region through the Foundry quota process, separately from Azure VM quota. ✓

C Increasing Search storage quota grants the required accelerator allocation.

D An API key removes the need for the managed-compute quota allocation.

Why the correct answer is right

The captured new Foundry managed-compute guide explicitly separates its quota from Azure VM quota and scopes it by accelerator SKU and region. The older classic VM-based recipe must not be substituted for this generation.

Why the other options are wrong

- **A:** The guide states that existing Azure VM quota cannot be applied to this managed-compute deployment.
- **C:** Search storage is a different resource dimension and does not supply accelerator quota.
- **D:** Authentication credentials do not bypass the service's capacity-allocation requirement.

AI103-A3.01

Sources

S004: Deployment overview for Microsoft Foundry Models
https://learn.microsoft.com/azure/foundry/concepts/deployments-overview
Snapshot SHA-256: 91520b0f84e6e2823bb27945cb8e25f716440b7680ffc7ea9689d8025f1d080f

Q15 Domain A: Plan and manage an Azure AI solution | single choice | medium

An agent team wants to correlate poor answers with evaluation runs. Tool arguments may contain account identifiers and an access token. Which telemetry policy best preserves diagnostic value while following the captured tracing guidance?

A Record every raw argument and credential so any operation can be reproduced exactly.

B Remove all trace identifiers, timestamps and evaluation IDs before exporting any data.

C Export raw spans first and rely entirely on a nightly script to remove secrets later.

D Keep consistent operation and evaluation identifiers, redact sensitive fields before export, and apply log-equivalent access and retention controls. ✓

Why the correct answer is right

The tracing guide recommends consistent attributes and evaluation-run correlation while removing or masking personal data, secrets and credentials before telemetry storage. Trace data needs appropriate access and retention controls.

Why the other options are wrong

- **A:** The guidance explicitly excludes secrets and credentials from prompts, tool arguments and span attributes retained in telemetry.
- **B:** Removing useful nonsecret correlation fields prevents the intended quality/performance analysis and is not required by the redaction guidance.
- **C:** Redacting only after export leaves sensitive material in telemetry before the cleanup runs, contrary to the stated before-storage guidance.

AI103-A4.03

Sources

S013: Agent tracing overview
https://learn.microsoft.com/azure/foundry/observability/concepts/trace-agent-concept
Snapshot SHA-256: 895295a4b74d7cabaa0daf62bd6cd5239daf25b6ea85a2268dc2cdf8483b19e8

Q16 Domain C: Implement computer vision solutions multi select medium

An app uses the captured Azure Content Safety image:analyze API contract (2024-09-01). Its policy checks all four harm categories, rejects an image when any category is Medium or High, and otherwise allows it. Which TWO implementation choices match that policy and the documented image API?

A Allow any image when the API returns HTTP success, because successful analysis means the content is approved.

B Reject when any returned category has severity 4 or 6; otherwise allow the image under the stated policy. ✓

C Request EightSeverityLevels for images and reject only severity values 5, 6 or 7.

D Request only the Violence category, because it includes the other three harm categories.

E Analyze all four categories using FourSeverityLevels, whose documented values are 0, 2, 4 and 6. ✓

Why the correct answer is right

The documented image moderation contract supports FourSeverityLevels: 0 Safe, 2 Low, 4 Medium and 6 High. Analyzing all four categories and rejecting any result at 4 or 6 implements the stated rule. HTTP success indicates successful analysis; the app still applies its acceptance policy.

Why the other options are wrong

- **A:** Successful analysis is not itself a content-approval decision. The result supplies category severities, and acceptance depends on the configured policy.
- **C:** The captured image API supports FourSeverityLevels, not the proposed eight-level output. The proposed threshold also fails to follow the stated Medium-or-High rule for the documented values.
- **D:** Hate, SelfHarm, Sexual and Violence are separate categories. Requesting only Violence omits signals that the policy explicitly requires.

AI103-C3.01

Sources

S042: QuickStart: Analyze image content
https://learn.microsoft.com/azure/ai-services/content-safety/quickstart-image
Snapshot SHA-256: 6a40b6fdabe9e2a91838a3026f8530d0229c70acd401c4fab094a6c0c63b8670

Q17 Domain C: Implement computer vision solutions multi select easy

A Sora 2 preview v1 remix call returns a new video ID with status queued. Which TWO actions correctly continue the asynchronous workflow to obtain the edited media?

A Track the job for the newly returned video ID until it completes successfully. ✓

B Download the original video ID immediately because remix overwrites its media in place.

C Treat the queued JSON response itself as the completed video file.

D After successful completion, retrieve the content associated with the new video ID. ✓

E Submit the same remix repeatedly until a call returns a binary video instead of a job.

Why the correct answer is right

The remix starts a new asynchronous video job. Track that returned job and retrieve its resulting media after completion. A queued acknowledgment is not the finished media.

Why the other options are wrong

- **B:** The returned new ID identifies the requested remix; using the original ID retrieves the original media rather than the new result.
- **C:** The response contains job metadata, not the finished media payload.
- **E:** Repeated submissions create additional work rather than following the documented status-and-retrieval lifecycle.

AI103-C1.04

Sources

S018: Video generation with Sora 2 (preview)
https://learn.microsoft.com/azure/foundry/openai/concepts/video-generation
Snapshot SHA-256: 32257da44487ca802a87e1ebad7ef31dc91f45f034d3334fb60b1c7d53ffb808

Q18 Domain C: Implement computer vision solutions multi select medium

An AI assistant proposes alt text for a webpage. A magnifying-glass image is the only label on a search button, and a decorative wave adds no information. Which TWO recommendations follow the captured W3C decision tree?

A Describe the search icon's exact color and pixel dimensions instead of naming its action.

B Give the search button an accessible text alternative that communicates its search action. ✓

C Use the decorative wave's filename as its alt text.

D Omit the alt attribute on the decorative image and rely on the browser to choose a description.

E Use an empty alt attribute for the purely decorative wave. ✓

Why the correct answer is right

Functional images need text that communicates the action or destination. Purely decorative images use an empty alt attribute so they do not add unnecessary spoken content.

Why the other options are wrong

- **A:** Appearance alone does not communicate what activating the button does.
- **C:** A filename adds irrelevant information for an image with no informative purpose.
- **D:** The decision tree calls for an empty alt attribute, which is an explicit choice rather than an absent attribute.

AI103-C2.04

Sources

S050: W3C WAI An alt Decision Tree
https://www.w3.org/WAI/tutorials/images/decision-tree/
Snapshot SHA-256: fa71070e3609f8825849fe94c3b718fb8632724b4c88ed9f3e5ea54615f36404

Q19 Domain D: Implement text analysis solutions multi select medium

A speech-to-text application consistently misrecognizes specialized equipment terms. The team has representative domain recordings and transcriptions. Which TWO activities match the documented Custom Speech workflow?

A Use the domain data to test and train a custom recognition model, comparing its accuracy with a baseline. ✓

B Select a different text-to-speech voice as the sole repair to the speech recognizer.

C Deploy the selected custom recognition model to a custom endpoint for the application. ✓

D Replace spoken-audio recognition with a model that only ranks search documents.

E Assume customization guarantees perfect recognition and remove accuracy evaluation.

Why the correct answer is right

Custom Speech supports uploading domain data, testing and training recognition models, comparing accuracy and deploying a model to a custom endpoint. Those steps address recognition quality for the application.

Why the other options are wrong

- **B:** A synthesis voice controls generated speech, not how the input audio is transcribed.
- **D:** Search ranking is a different task and does not replace the required speech-recognition model.
- **E:** The documented workflow includes testing and comparing accuracy; training does not guarantee error-free recognition.

AI103-D2.02

Sources

S034: What is custom speech?
https://learn.microsoft.com/azure/ai-services/speech-service/custom-speech-overview
Snapshot SHA-256: 8bde6391b7bdee42b5e52ea8c995024dcda82d1a0b56acbdfb817207b2a8d511

Q20 Domain D: Implement text analysis solutions multi select medium

A Python Speech SDK application must recognize Italian speech, translate it into German, and later speak the translated text. Recognition and synthesis credentials are already available. Which TWO statements correctly describe this flow?

A Set the recognition source language to de-DE because German is the output language.

B Set speech_recognition_language to it-IT and add de as a translation target. ✓

C Read result.translations["it-IT"] as the German target output.

D Read the German translated text from the de entry after successful translation, then send that text to the synthesis step. ✓

E Treat configuring a translation target as proof that a separate SpeechSynthesizer has already produced audio.

Why the correct answer is right

The source locale identifies the incoming speech, while a target language selects the translation. The translations dictionary is keyed by target language. Generating spoken output is a further synthesis operation.

Why the other options are wrong

- **A:** German is the target, not the language being recognized from the incoming Italian audio.
- **C:** The requested target is de; it-IT is the recognition locale and is not the configured German translation key.
- **E:** A configured target or a text translation does not establish that the separate synthesis step has produced audio.

AI103-D2.04

Sources

S046: How to recognize and translate speech
https://learn.microsoft.com/azure/ai-services/speech-service/how-to-translate-speech
Snapshot SHA-256: 8a73fb1e63f49dde972815b860758a57ff3d620842220f9e29200bccf573c39e

Q21 Domain E: Implement information extraction solutions sequence order medium

An existing Search pipeline has an index and an embedding deployment configured. The source documents are too long for one embedding request. Arrange the four data-processing steps in dependency order. Configuration setup is complete; these are the operations on each document.

Correct order:

1. Retrieve the raw document content from the supported data source.
2. Split the retrieved text into chunks within the embedding model's input limits.
3. Generate an embedding vector for each prepared text chunk.
4. Write the resulting chunk text and vectors into the configured search index.

Why the correct answer is right

The indexer retrieves source content, chunking prepares units small enough for embedding, the embedding skill produces vectors, and the index receives the processed content. Keeping text as well as vectors supports readable retrieval results.

Answer and step notes

- **A:** Chunking follows content retrieval and precedes embedding for these oversized documents.
- **B:** The index receives the generated text/vector records after those records exist.
- **C:** The raw input must be retrieved before the downstream operations can process it.
- **D:** The model embeds the prepared chunks; writing finished vector records depends on that output.

AI103-E1.01

Sources

S026: Integrated vector embedding in Azure AI Search
https://learn.microsoft.com/azure/search/vector-search-integrated-vectorization
Snapshot SHA-256: 45a6b445fbe4f7f65334693eb96b2de7b021e0b50ce46b363ab47c5de0a11f0d

Q22 Domain E: Implement information extraction solutions sequence order easy

A document analyzer is already configured in Content Understanding. Using the captured Python SDK 1.1.0 workflow, arrange these application steps to process one document. Authentication and model configuration are complete.

Correct order:

1. Begin analysis with the configured analyzer and document input.
2. Wait for the returned long-running operation to complete, such as by using the poller's result() method.
3. Read the completed analysis result's structured content and markdown.
4. Pass the extracted representation to the downstream application.

Why the correct answer is right

Analysis is asynchronous. Begin the operation, wait for completion, extract the completed content, then use that content downstream. The initial operation acknowledgment is not the final analysis result.

Answer and step notes

- **A:** Structured content is processed after the long-running analysis completes.
- **B:** Beginning analysis creates the operation that can subsequently be polled.
- **C:** The downstream application consumes the extracted result, so this step follows result processing.
- **D:** The SDK poller handles waiting for the asynchronous operation before the completed result is consumed.

AI103-E2.03

Sources

S021: Azure AI Content Understanding client library for Python - version 1.1.0
https://learn.microsoft.com/python/api/overview/azure/ai-contentunderstanding-readme?view=azure-python
Snapshot SHA-256: aa5fd0abb99fe608e62d3909ffd7bc4eac9904dc931f427c9e3788351fbc7b42

Q23 Domain A: Plan and manage an Azure AI solution matching magnet medium

Match each hosted-agent failure situation to the most appropriate listed next action. Required session state must be retained, moving regions is prohibited, and the model token quota is not exhausted. Use each selected action once; one action is unused.

Correct matches:

session_quota_exceeded; several sessions no longer need active compute, but their stored state is still required.	**→ Stop sessions that no longer need active compute.**
regional_session_quota_exceeded occurs intermittently; the client currently retries immediately with no delay.	**→ Apply exponential backoff with jitter.**
regional_session_quota_exceeded persists after appropriate retry/backoff in the approved region.	**→ Create an Azure support request.**

Choices: Apply exponential backoff with jitter., Increase the model tokens-per-minute quota., Create an Azure support request., Stop sessions that no longer need active compute.

Why the correct answer is right

The guide distinguishes the concurrent-session layer from regional session capacity and separates initial retry behavior from persistent-error escalation. Stopping unnecessary active compute preserves required state; deleting all stored sessions would not.

Answer and step notes

- **A:** Stopping sessions addresses unneeded active compute without choosing the forbidden deletion of required state.
- **B:** Backoff with jitter is an initial response to intermittent regional capacity failures when moving regions is unavailable.
- **C:** The guidance directs support escalation when the regional error persists. Raising a model token quota is unused because it addresses a different limit.

AI103-A3.01

Sources

S009: Foundry Agent Service limits, quotas, and regional support
https://learn.microsoft.com/azure/foundry/agents/concepts/limits-quotas-regions
Snapshot SHA-256: d83fd890729d322d4ffa580ca5a8db2bed5bcd783dae47f5da2b05985f4b3041

Q24 Domain D: Implement text analysis solutions matching magnet easy

Choose the most targeted Azure Language capability for each text-processing requirement. Use each selected capability once; one choice is unused.

Correct matches:

Identify personal identifiers in submitted text so the application can route them to a redaction workflow.	**→ PII detection**
Determine whether incoming text is primarily French, German or another supported language.	**→ Language detection**
Label mentions in text using predefined entity types such as organizations and locations.	**→ Prebuilt named entity recognition**

Choices: Summarization, Language detection, PII detection, Prebuilt named entity recognition

Why the correct answer is right

The captured Language overview distinguishes PII detection, language detection and prebuilt NER. Each is targeted at the respective requirement, while a general document summary would not supply those specialized results.

Answer and step notes

- **A:** PII detection identifies entities associated with individuals; subsequent redaction policy remains an application decision.
- **B:** Language detection evaluates the input language and dialect rather than classifying named entities.
- **C:** Prebuilt NER assigns predefined entity types. Summarization is unused because none of these tasks asks for a condensed account of the document.

AI103-D1.01 AI103-D1.02

Sources

S030: What is Azure Language in Foundry Tools?
https://learn.microsoft.com/azure/ai-services/language-service/overview
Snapshot SHA-256: debd3c5f132c72a0d258daa4d82d950834da5b4c89e3b42b20cf32982a169e1d

Q25 Domain E: Implement information extraction solutions | code or config artifact | medium

A blob indexer and OCR skillset already exist. The OCR skill consumes /document/normalized_images/* and names its text output text. The target index has a searchable ocr_text field. The fragment below fails to supply OCR input and treats enriched output as a source field. Which paired correction addresses both problems?

Indexer fragment (illustrative JSON; other settings omitted)

```
{
  "parameters": {
    "configuration": {"dataToExtract": "contentAndMetadata",
                      "imageAction": "none"}
  },
  "fieldMappings": [
    {"sourceFieldName": "/document/normalized_images/*/text",
     "targetFieldName": "ocr_text"}
  ]
}
```

A Keep imageAction as none and rename fieldMappings to vectorQueries.

B Set imageAction to generateNormalizedImages and use outputFieldMappings for the enriched OCR text path. ✓

C Set imageAction to generateNormalizedImages but keep fieldMappings for the enriched OCR text path.

D Keep imageAction as none and use outputFieldMappings, relying on mapping alone to create image inputs.

Why the correct answer is right

The OCR skill needs normalized-image inputs generated by the indexer imageAction setting. Its text is an enriched document-tree node, so outputFieldMappings carries that node into the index field. Both input generation and the output mapping must be correct.

Why the other options are wrong

- **A:** Disabling image generation still leaves the OCR input absent, and vectorQueries is not the indexer mapping mechanism for enriched output.
- **C:** This produces normalized images but still uses the source-field mapping mechanism for an enriched node.
- **D:** An output mapping moves an existing enriched value; it does not generate the normalized-image inputs required by OCR.

AI103-E1.04

Sources

S043: OCR cognitive skill
https://learn.microsoft.com/azure/search/cognitive-search-skill-ocr
Snapshot SHA-256: 0a5cdd696dcbae744d32db7fd89a69926f0203462f86d6dc25793ab07d001b3d

S028: Skillset concepts in Azure AI Search
https://learn.microsoft.com/azure/search/cognitive-search-working-with-skillsets
Snapshot SHA-256: 3d24961260a3a7ca04165638301622270 4c25dec2dff53743bbaef628236312b

Quick Answer Key

#	Type	Correct Answer
Q1	case study question	**C**
Q2	case study question	**A**
Q3	case study question	**B**
Q4	case study question	**D**
Q5	dropdown completion	**required → ["order_id", "note"] | additional → false | format → text.format**
Q6	log or artifact interpretation	**C**
Q7	single choice	**A**
Q8	single choice	**D**
Q9	single choice	**B**
Q10	single choice	**C**
Q11	single choice	**D**
Q12	single choice	**A**
Q13	single choice	**C**
Q14	single choice	**B**
Q15	single choice	**D**
Q16	multi select	**B, E**
Q17	multi select	**A, D**
Q18	multi select	**B, E**
Q19	multi select	**A, C**
Q20	multi select	**B, D**
Q21	sequence order	**1. C → 2. A → 3. D → 4. B**
Q22	sequence order	**1. B → 2. D → 3. A → 4. C**
Q23	matching magnet	**A → Stop sessions that no longer need active compute. | B → Apply exponential backoff with jitter. | C → Create an Azure support request.**
Q24	matching magnet	**A → PII detection | B → Language detection | C → Prebuilt named entity recognition**
Q25	code or config artifact	**B**